

10b-more-json-practice-with-nobelprize-org

Introduction

This assignment uses the public Nobel Prize API (v1) at <http://api.nobelprize.org/v1/>. Two JSON endpoints are consumed:

- `prize.json` — one record per prize (year + category) with nested laureates
- `laureate.json` — one record per laureate with nested prizes and affiliations

Both are pulled, flattened into tidy data frames, and used to answer four questions below.

```
library(tidyverse)
library(jsonlite)

# Pull both endpoints once
prizes_raw    <- fromJSON("http://api.nobelprize.org/v1/prize.json",
                          simplifyVector = FALSE)
laureates_raw <- fromJSON("http://api.nobelprize.org/v1/laureate.json",
                          simplifyVector = FALSE)
```

Tidy the prize data

Each prize has a list of laureates, so we unnest it into one row per laureate-prize.

```
prizes_df <- tibble(prize = prizes_raw$prizes) |>
  unnest_wider(prize) |>
  mutate(year = as.integer(year)) |>
  unnest_longer(laureates) |>      # one row per laureate on each prize
  unnest_wider(laureates) |>
  filter(!is.na(id))              # drop years with no laureates (e.g., war years)

glimpse(prizes_df)
```

```

Rows: 1,026
Columns: 8
$ year          <int> 2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025~
$ category      <chr> "chemistry", "chemistry", "chemistry", "economics", ~
$ id            <chr> "1053", "1054", "1055", "1058", "1059", "1060", "105~
$ firstname     <chr> "Susumu", "Richard", "Omar M.", "Joel", "Philippe", ~
$ surname       <chr> "Kitagawa", "Robson", "Yaghi", "Mokyr", "Aghion", "H~
$ motivation    <chr> "\"for the development of metal-organic frameworks\"~
$ share        <chr> "3", "3", "3", "2", "4", "4", "1", "1", "3", "3", "3~
$ overallMotivation <chr> NA, NA, NA, "\"for having explained innovation-
drive~

```

Tidy the laureate data

```

laureates_df <- tibble(l = laureates_raw$laureates) |>
  unnest_wider(l) |>
  mutate(id = as.character(id))

# One row per (laureate, prize) with affiliation info for Q4
lp_df <- laureates_df |>
  select(id, firstname, surname, gender, bornCountry, prizes) |>
  unnest_longer(prizes) |>
  unnest_wider(prizes, names_sep = "_") |>
  mutate(prize_year = as.integer(prizes_year))

glimpse(lp_df)

```

```

Rows: 1,026
Columns: 12
$ id            <chr> "1", "2", "3", "4", "5", "6", "6", "8", "9", ~
$ firstname     <chr> "Wilhelm Conrad", "Hendrik A.", "Pieter", "He~
$ surname       <chr> "Röntgen", "Lorentz", "Zeeman", "Becquerel", ~
$ gender        <chr> "male", "male", "male", "male", "male", "fema~
$ bornCountry   <chr> "Prussia (now Germany)", "the Netherlands", "~
$ prizes_year   <chr> "1901", "1902", "1902", "1903", "1903", "1903~
$ prizes_category <chr> "physics", "physics", "physics", "physics", "~
$ prizes_share  <chr> "1", "2", "2", "2", "4", "4", "1", "1", "1", ~
$ prizes_motivation <chr> "\"in recognition of the extraordinary servic~
$ prizes_affiliations <list> [["Munich University", "Munich", "Germany"]~
$ prizes_overallMotivation <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ prize_year    <int> 1901, 1902, 1902, 1903, 1903, 1903, 1911, 190~

```

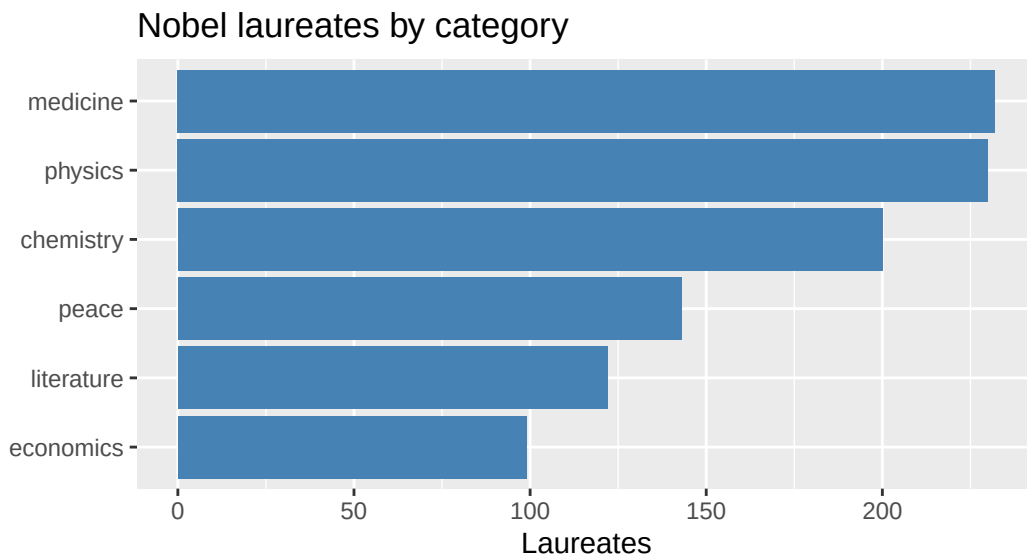
Question 1: Which category has awarded the most Nobel Prizes?

A simple count of laureates by category.

```
q1 <- prizes_df |>
  count(category, sort = TRUE)
q1
```

```
# A tibble: 6 x 2
  category      n
  <chr>    <int>
1 medicine   232
2 physics   230
3 chemistry  200
4 peace     143
5 literature 122
6 economics  99
```

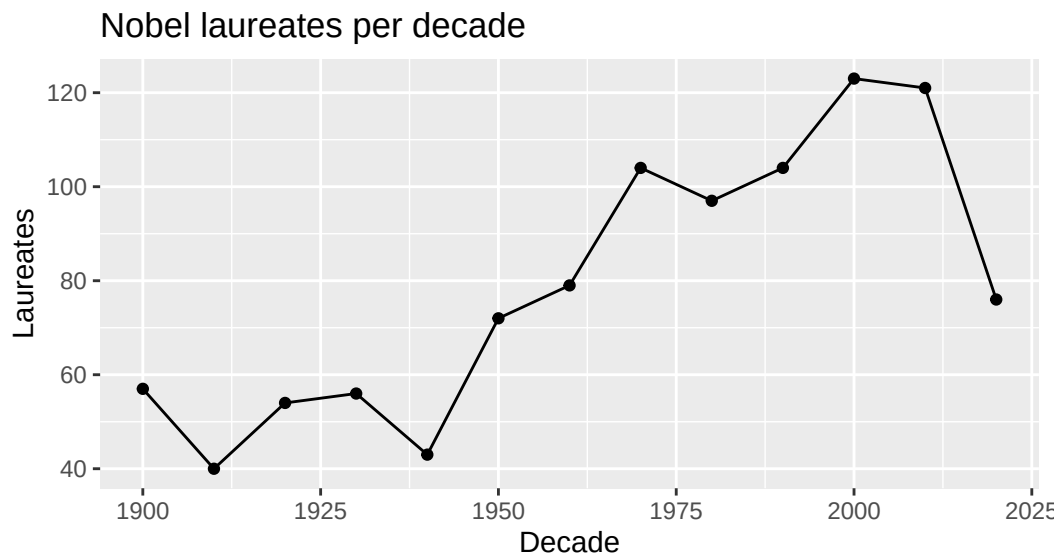
```
ggplot(q1, aes(reorder(category, n), n)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(x = NULL, y = "Laureates", title = "Nobel laureates by category")
```



Question 2: How has the number of laureates per decade changed over time?

```
q2 <- prizes_df |>
  mutate(decade = (year %/% 10) * 10) |>
  count(decade)

ggplot(q2, aes(decade, n)) +
  geom_line() + geom_point() +
  labs(x = "Decade", y = "Laureates",
       title = "Nobel laureates per decade")
```



q2

```
# A tibble: 13 x 2
  decade    n
  <dbl> <int>
1  1900     57
2  1910     40
3  1920     54
4  1930     56
5  1940     43
```

6	1950	72
7	1960	79
8	1970	104
9	1980	97
10	1990	104
11	2000	123
12	2010	121
13	2020	76

The count rises over time, reflecting the modern practice of splitting prizes among multiple laureates.

Question 3: What is the gender distribution within each category?

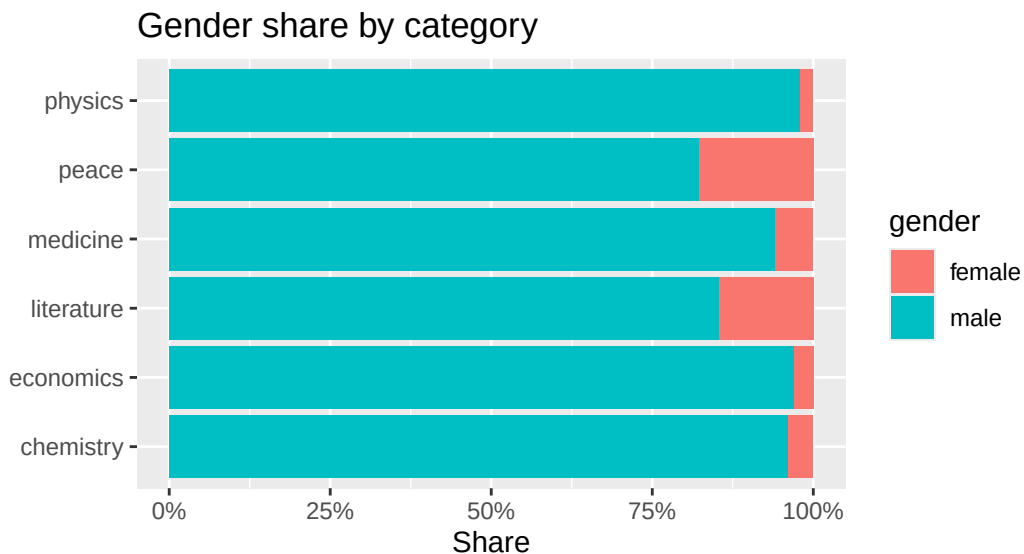
This joins laureate-level gender with prize-level category.

```
# Join gender from laureates_df onto prizes_df by laureate id
q3 <- prizes_df |>
  mutate(id = as.character(id)) |>
  left_join(select(laureates_df, id, gender), by = "id") |>
  filter(gender %in% c("male", "female")) |>      # drop organizations
  count(category, gender) |>
  group_by(category) |>
  mutate(pct_female = round(100 * n[gender == "female"] / sum(n), 1)) |>
  ungroup()
q3
```

```
# A tibble: 12 x 4
  category gender      n pct_female
  <chr>    <chr> <int>    <dbl>
1 chemistry female      8         4
2 chemistry male    192         4
3 economics female      3         3
4 economics male     96         3
5 literature female    18        14.8
6 literature male   104        14.8
7 medicine  female    14         6
8 medicine  male   218         6
9 peace     female    20        17.9
```

10	peace	male	92	17.9
11	physics	female	5	2.2
12	physics	male	225	2.2

```
ggplot(q3, aes(category, n, fill = gender)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(x = NULL, y = "Share", title = "Gender share by category") +
  coord_flip()
```



Peace and Literature have the highest share of female laureates; Physics and Economics have the lowest.

Question 4: Which country has “lost” the most laureates?

The advanced question: find laureates **born** in one country but **affiliated** with a different country at the time of the award. This requires unnesting prize affiliations and comparing against born country.

```
# Unnest affiliations inside each prize
brain_drain <- lp_df |>
  select(id, bornCountry, prize_year, prizes_category, prizes_affiliations) |>
```

```

unnest_longer(prizes_affiliations) |>
unnest_wider(prizes_affiliations, names_sep = "_") |>
rename(affil_country = prizes_affiliations_country) |>
filter(!is.na(bornCountry),
       !is.na(affil_country),
       nzchar(bornCountry),
       nzchar(affil_country),
       bornCountry != affil_country) |>
distinct(id, bornCountry, affil_country)      # one row per laureate move

q4 <- brain_drain |>
  count(bornCountry, sort = TRUE) |>
  slice_head(n = 10)
q4

```

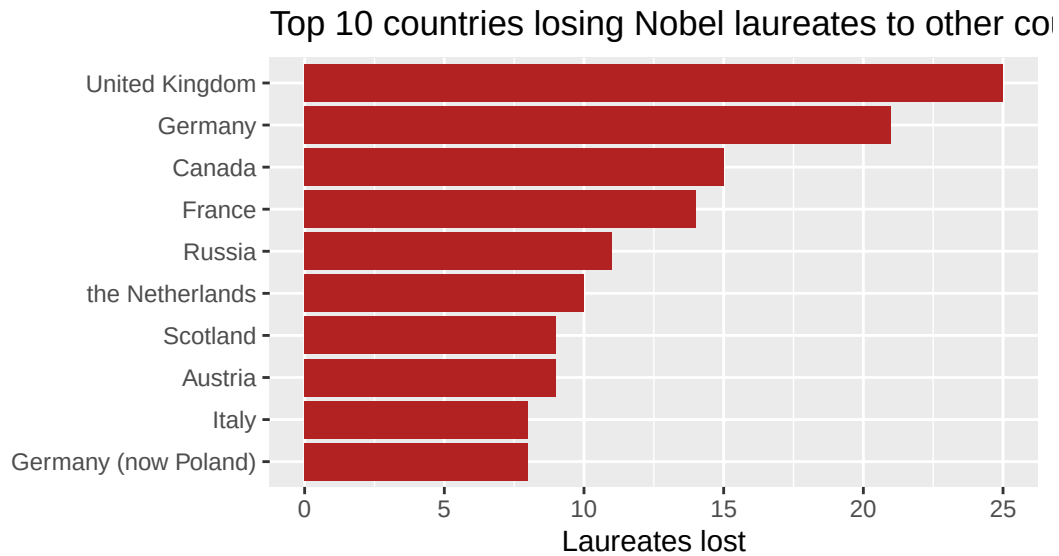
A tibble: 10 x 2

	bornCountry	n
	<chr>	<int>
1	United Kingdom	25
2	Germany	21
3	Canada	15
4	France	14
5	Russia	11
6	the Netherlands	10
7	Austria	9
8	Scotland	9
9	Germany (now Poland)	8
10	Italy	8

```

ggplot(q4, aes(reorder(bornCountry, n), n)) +
  geom_col(fill = "firebrick") +
  coord_flip() +
  labs(x = NULL, y = "Laureates lost",
       title = "Top 10 countries losing Nobel laureates to other countries")

```



Germany (Germany and Germany (now Poland)) tops the list — largely because many scientists born in the pre-war German Empire were affiliated with U.S. or U.K. institutions by the time they won, reflecting the 20th-century migration of scientific talent.

Summary

#	Question	Answer
1	Most-awarded category	Medicine / Physics (largest counts)
2	Laureates per decade	Rising trend, peaking in recent decades
3	Gender balance	Peace & Literature most balanced; Physics least
4	Biggest “brain drain”	Germany